

Advancing Acoustic Droplet Vaporization for Tissue Characterization Using Quantitative Ultrasound and Transfer Learning

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Abstract—Acoustic droplet vaporization (ADV) is an emerging technique with expanding applications in biomedical ultrasound. ADV-generated bubbles can function as microscale probes that provide insights into the mechanical properties of their surrounding microenvironment. This study investigated the acoustic and imaging characteristics of phase-shift nanodroplets in fibrin-based, tissue-mimicking hydrogels using passive cavitation detection and active imaging techniques, including B-mode and contrast-enhanced ultrasound. The findings demonstrated that the backscattered signal intensities and pronounced nonlinear acoustic responses, including subharmonic and higher harmonic frequencies, of ADV-generated bubbles correlated inversely with fibrin density. Additionally, we quantified the mean echo intensity, bubble cloud area, and second-order texture features of the generated ADV bubbles across varying fibrin densities. ADV bubbles in softer hydrogels displayed significantly higher mean echo intensities, larger bubble cloud areas, and more heterogeneous textures. In contrast, texture uniformity, characterized by variance, homogeneity, and energy, correlated directly with fibrin density. Furthermore, we incorporated transfer learning with convolutional neural networks, adapting AlexNet into two specialized models for differentiating fibrin hydrogels. The integration of deep learning techniques with ADV offers great potential, paving the way for future advancements in biomedical diagnostics.

Index Terms—Acoustic droplet vaporization, tissue characterization, transfer learning, phase-shift droplets, ultrasound.

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I. INTRODUCTION

PATHOLOGICAL processes frequently alter the structural and mechanical properties of soft tissues, serving as key biomarkers for early disease detection [1]. For instance, liver stiffness correlates with the stage of fibrosis, where increased stiffness indicates advanced disease progression [2]. Similarly, thrombus stiffness increases with age due to fibrin crosslinking and replacement of cellular components with collagen deposits [3]. These mechanical alterations in soft tissues can be detected by clinicians through tissue characterization, making the assessment of tissue biomechanics critical not only for understanding overall tissue health but also for monitoring disease progression. Consequently, there is a growing interest in non-invasive methods to characterize the mechanical properties of soft tissues, which range broadly, from tens of Pa to tens of kPa [4]. This necessitates the development of sensitive and accurate assessment techniques. Ultrasound is a desirable tool for this purpose due to its many inherent advantages, such as wide availability, non-invasiveness, cost-effectiveness, and real-time capabilities. Ultrasound elastography techniques are commonly used in clinical diagnostics predominantly for conditions such as liver fibrosis, breast lesions, and thyroid nodules [5], [6]. However, there are several limitations associated with ultrasound elastography including operator skill dependence, leading to low reproducibility, limited dynamic range for probing tissue elasticity, and tissue size requirements [7], [8]. Additionally, these methods assume homogeneous tissue properties and lack the spatial resolution required for microscale characterization to detect early-stage pathology.

Acoustic droplet vaporization (ADV), with a unique ability to generate bubbles in situ—only where and when needed [9], [10], has emerged as a promising platform to address clinical challenges. In recent years, the applications of ADV have expanded to new areas such as oxygen scavenging [11], modulation of biomaterial properties [12], [13], and ultrasound localization microscopy [14]. Phase-shift droplets used in ADV can be tailored for specific applications by adjusting parameters such as the volatility of the perfluorocarbon liquid core, droplet size distribution, and ultrasound parameters used to trigger them [15], [16], [17]. Bubbles formed by ADV allow precise and localized probing of the surrounding medium. Despite the potential, most research on the effect of medium rheology on bubble

dynamics and acoustic response have been primarily numerical, thus experimental research in this area remains limited.

The increasing use of phase-shift droplets in biomedical applications necessitates comprehensive optical and acoustic characterizations. Optical techniques, such as high-speed microscopy, have been instrumental in studying bubble dynamics in an ultrasound field, providing critical insights into how parameters like excitation frequency [18], [19], perfluorocarbon core [18], [19], [20], and droplet size [21] influence the ADV bubble response. Recent studies utilizing high-speed microscopy have also demonstrated an inverse relationship between the elastic modulus of the surrounding medium and key ADV bubble dynamics, including the maximum expansion ratio and collapse time [18], [22].

While optical characterization methods provide valuable insights into bubble dynamics in transparent media with a highly diluted droplet suspension, their application in vivo and in clinical settings is limited. Acoustic characterization of phase-shift droplets, on the other hand, provides more practical clinical relevance. In acoustic experiments, scattered emissions are measured instead of radial responses. In their liquid phase, phase-shift droplets are poor scatterers of ultrasound [23]; however, upon vaporization, the resultant gas bubbles become highly efficient scatterers. Both linear (fundamental (f_0)) and nonlinear (f_0/n_i and $n_i f_0$) frequency components of the scattered responses have been shown to significantly increase during ADV [24], [25]. The harmonic generation of ADV bubbles has significant implications not only for contrast imaging but also for imaged-guided therapeutic interventions. Although the effects of excitation frequency (0.3–40 MHz), pulse duration (μ s–ms range), droplet size (0.1–30 μ m), and perfluorocarbon liquid core (C_3F_8 – C_9F_{20}) on the ADV threshold have been extensively studied, the stability of the generated bubbles is an area of ongoing research. Time-lapse confocal microscopy has shown that bubbles generated from droplets with perfluoropentane (C_5F_{12}) and perfluorohexane (C_6F_{14}) cores remain stable for minutes post-ADV [18]. The slow dissolution of gases with lower Ostwald coefficients, such as perfluorocarbons, combined with the inverse relationship between diffusion rate and molecular weight of the gas, contributes to the prolonged stability of these bubbles.

In this study, we characterized the acoustic emissions of phase-shift nanodroplets embedded in media with varying elastic moduli, using a combination of passive cavitation detection and active imaging methods, including B-mode and contrast-enhanced ultrasound (CEUS) imaging. Hydrogels with varying fibrin concentrations were used to mimic the soft tissue environment, providing a controlled setting to assess the relationship between elastic modulus and acoustic emissions. Perfluoropentane nanodroplets were selected due to their optimal size for applications involving soft tissue penetration and accumulation. We analyzed the nonlinearity and stability of the backscattered emissions from ADV bubbles both in time and frequency domains via passive cavitation detection. Additionally, the morphological and textural features of ADV bubble clouds were analyzed in B-mode and nonlinear CEUS images to assess the effect of fibrin concentration. Such textural analysis of ADV bubble

clouds has not been studied before. Finally, we incorporated transfer learning with convolutional neural networks to enhance the diagnostic potential of ADV in tissue characterization. The integration of ADV with deep learning frameworks offers a nuanced method for advancing non-invasive tissue diagnostics and expanding the applications of ADV.

II. MATERIALS AND METHODS

A. Phase-Shift Nanodroplet Preparation and Characterization

Phase-shift nanodroplets were synthesized using a thin-film hydration method, following a previously established protocol [14]. The nanodroplets consisted of a liquid perfluoropentane (PFP, Strem Chemicals, Newburyport, MA, USA) core encapsulated within a stabilizing lipid shell. The precursor lipid film, dissolved in chloroform at a total concentration of 1 mg/mL, was comprised of 1,2-dipalmitoyl-sn-glycero-3-phosphocholine (DPPC, Avanti Polar Lipids, Alabaster, AL, USA), 1,2-dipalmitoyl-sn-glycero-3-phosphate (sodium salt) (DPPA, Avanti Polar Lipids), and 1,2-dipalmitoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(polyethylene glycol)-5000] (PEG5000PE, Avanti Polar Lipids) in a molar ratio of 9:1:1. Following dissolution, the chloroform was completely evaporated in a vacuum chamber (Isotemp vacuum oven, Model 282A, Fisher Scientific, Dubuque, IA, USA), leaving a thin lipid film. The dried lipid film was then rehydrated with an aqueous buffer solution consisting of 80% (v/v) phosphate-buffered saline (PBS), 10% (v/v) propylene glycol (Sigma-Aldrich), and 10% (v/v) glycerol. To produce the nanodroplets, 750 μ L of the lipid solution was combined with 80 mg of liquid PFP, and the mixture was sonicated (Branson 450A, Emerson, St. Louis, MO, USA) for 30 seconds on ice. The resulting nanodroplet suspension was then stored in the fridge for three days, allowing larger droplets to settle. The resulting supernatant was used for all experiments. Nanodroplets were characterized via ZetaView (Particle Metrix Inc., Ammersee, Germany) at room temperature. The average concentration of the nanodroplets was 2.3×10^9 particles/mL, with a mean diameter of 327.5 ± 71.5 nm.

B. Preparation of Tissue Mimicking Fibrin-Based Hydrogels

Tissue-mimicking hydrogels were prepared as previously described [18], [22], [26]. Briefly, a stock solution of bovine fibrinogen (Sigma-Aldrich) was prepared at a concentration of 4.5% (w/v) in FluoroBrite Dulbecco's Modified Eagle Medium (DMEM, Life Technologies). The hydrogels were formed by combining the fibrinogen solution, DMEM, 0.1% (v/v) nanodroplets, and human recombinant thrombin (2 U/mL, Re-cothrom, Baxter, Deerfield, IL, USA). The thrombin catalyzed the polymerization of fibrinogen into fibrin, forming a stable hydrogel matrix. To investigate the effect of varying fibrin concentrations on the acoustic emissions of nanodroplets, hydrogels with final fibrin concentrations ranging from 0.2% to 4% (w/v) were prepared. Nanodroplets suspended in DI water without fibrin (i.e., 0% fibrin) were used to represent a medium

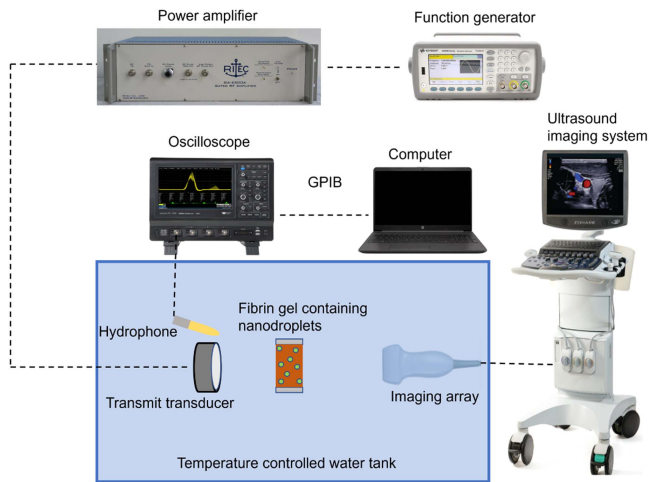


Fig. 1. Schematic representation of the experimental setup. A calibrated, focused transducer was used to generate acoustic droplet vaporization (ADV) of phase-shift nanodroplets embedded in fibrin-based hydrogels with varying Young's moduli. The resulting backscattered emissions from ADV were passively captured using a hydrophone. Additionally, ADV-induced bubble clouds were visualized using plane-wave imaging in an active mode.

with zero elasticity. Additionally, blank hydrogels (i.e., without nanodroplets) were prepared to monitor any potential cavitation effects arising from acoustic parameters used in ADV experiments. The hydrogel mixtures were polymerized in custom cylindrical molds made from polyvinyl chloride pipes (diameter: 20 mm, thickness: 7 mm), with both ends sealed using Tegaderm (Tegaderm, 3M Health Care) membranes, which are known to be acoustically transparent [26]. To assess the mechanical properties of the fibrin hydrogels, atomic force microscopy (AFM, FlexAFM, Nanosurf, Liestal, Switzerland) was used to measure the Young's modulus. For AFM analysis, fibrin hydrogels (height: ≈ 1 mm) were mounted on poly-L-lysine coated coverslips. A gold-coated AFM probe with a pyramidal tip (nominal diameter: 20 nm, PPP-CONTAuD, NanoAndMore, Watsonville, CA, USA) was used in contact mode to perform the mechanical indentation. The cantilever's spring constant was calibrated using the built-in thermal tuning method. The compressive Young's modulus was calculated from the force-indentation curves using the Derjaguin-Muller-Toporov (DMT) model in AtomicJ software (<http://sourceforge.net/projects/jrobust>) [13].

C. Ultrasound Setup and Data Acquisition

The experimental setup for generating ADV and recording the resulting acoustic emissions is schematically shown in Fig. 1. A calibrated, spherically focused transducer (H-108, Sonic Concepts) operating at 2.5 MHz was used to generate ADV. The pressure profile of the focal volume was characterized in a scanning tank (AIMS III, Onda, CA, USA) using a custom fiber optic hydrophone with a fiber diameter of 105 μm and sensitivity of 16.6 mV/MPa [27]. The peak negative pressure at the transducer focus was controlled by a waveform generator (33500B, Agilent Technologies) and amplified by a gated power amplifier (Ritec, GA-2500A). For all experiments, a 6-cycle

tone burst at a pulse repetition frequency of 100 Hz and a peak negative pressure of 3.2 MPa was used. The acoustic parameters were selected based on prior research [22], which demonstrated optimal sensitivity of ADV to variations in the elastic properties of the surrounding medium.

The transducer's geometric focus was aligned within the hydrogels, positioned approximately 3 mm behind the Tegaderm membrane, corresponding to the mid-thickness of the hydrogel samples. Acoustic emissions resulting from ADV were captured passively using a capsule hydrophone (HGL-1000, frequency range: 1–20 MHz, Onda, Sunnyvale, CA, USA). The recorded signals were acquired by a digital oscilloscope (sampling rate: 100 MHz, HDO4034, Teledyne LeCroy, Chestnut Ridge, NY, USA) using MATLAB (MathWorks, Natick, MA, USA) and processed offline as outlined previously [22]. The hydrophone was connected to a preamplifier and a DC block, as recommended by the manufacturer, and impedance-matched before connecting to the oscilloscope to minimize signal reflections and maintain signal integrity. For B-mode ultrasound imaging, a 12 MHz linear array probe (Mindray, L9-15) connected to a ZS3 diagnostic ultrasound system (Mindray North America, Mountain View, CA, USA) was used. CEUS images were acquired using a 6 MHz linear array (C2-9) connected to a Logiq E10 diagnostic ultrasound system (GE Healthcare, Mountain View, CA, USA) following prior publications [28], [29]. The probes were encased in a cover (Trojan Condoms; Church & Dwight Company, Ewing Township, NJ, USA) filled with coupling gel (EcoVue Ultrasound Gel, A Division of HR Pharmaceuticals, Inc., York, PA, USA) [30]. A computer-controlled three-axis positioning system ensured precise alignment of the hydrogels relative to the transmitting transducer, hydrophone, and linear array probes. To isolate and quantify vaporization signals generated from phase-shift droplets, B-mode and CEUS images were first acquired prior to transmitting the ADV pulse to capture the background signal. The differential between frames before and after ADV was computed. For each experimental condition, 3-second video sequences comprising a total of 64 frames were recorded. During data acquisition, the mechanical index (MI) was maintained at 0.3 for both B-mode and CEUS imaging, ensuring that no apparent ADV occurred within the hydrogels [26]. For each experimental condition, six measurements were collected for ultrasound imaging ($N = 6$). For hydrophone measurements, 20 radio-frequency (RF) signals were captured in sample mode, with nine repetitions for each condition ($N = 9$). All experiments were conducted in a tank filled with degassed ($< 25\%$ dissolved oxygen saturation) deionized (DI) water at 37 $^{\circ}\text{C}$.

D. Morphological and Textural Feature Analyses

Mean echo intensity was quantified, on log-decompressed images, by integrating pixel intensity values within a defined region of interest in each acquired frame. For area measurements, the threshold gray scale value separating the ADV bubble cloud from fibrin background was determined using Otsu's method [31]. To further analyze and compare the features in ultrasound images, we employed the gray level co-occurrence

matrix (GLCM) algorithm [32], a widely used method for quantifying second-order texture features [33], [34], [35]. First- and higher-order texture features have demonstrated high repeatability across variations in ultrasound scanners, beamformers, dynamic range settings, and the dimensions/shapes of regions of interest [36]. In this study, four commonly reported GLCM parameters—contrast, variance, homogeneity, and energy—were calculated to describe various aspects of the texture, including pixel intensity relationships and overall texture uniformity [37], [38], [39], [40].

- **Contrast:** This feature quantifies the intensity differences between a pixel and its neighbors across the entire image, capturing local variations in gray-level intensities.

$$\text{Contrast} = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} (i-j)^2 P(i,j)$$

where $P(i,j)$ is the normalized GLCM, defined as $\frac{C(i,j)}{\sum C(i,j)}$, and $C(i,j)$ represents the co-occurrence count of pixel intensity pairs.

- **Variance:** This parameter measures the spread of pixel intensity values around the mean, reflecting the overall variability within the texture.

$$\text{Variance} = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} (i-\mu)^2 \cdot P(i,j)$$

where μ represents the mean of the pixel intensity distribution in the GLCM.

- **Homogeneity:** Also referred to as the inverse difference moment, homogeneity evaluates the closeness of GLCM elements to the diagonal, providing insight into the uniformity of texture.

$$\text{Homogeneity} = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} \frac{P(i,j)}{1+|i-j|}$$

- **Energy:** This feature represents the sum of squared elements in the GLCM, which reflects the magnitude of pixel pair correlations and overall power in the texture.

$$\text{Energy} = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} P(i,j)^2$$

These textural features were computed using MATLAB's Image Processing Toolbox. The transducer-to-hydrogel angle was consistent across all the images.

E. AlexNet Modifications and Training

AlexNet is a well-established neural network architecture, known for its efficacy across various clinical applications [41], [42], [43]. The network consists of five convolutional layers followed by three fully connected layers, designed to extract and classify complex image features. In this study, we modified the original AlexNet architecture to develop two new models: BubbleNet1 and BubbleNet2, specifically designed for the classification of fibrin hydrogels and the prediction of elastic modulus. The overall workflow is schematically depicted in

Fig. 2. Two key modifications were made to adapt AlexNet into BubbleNet1 and BubbleNet2. First, the input layer was adjusted to accommodate single-channel images (e.g., scalogram, B-mode, or CEUS mode images) by reducing the input channels from three to one. The final B-mode and CEUS images were exported from the ultrasound scanners in log-compressed PNG format. Second, the number of output classes was reduced from 1000 to four, aligning with the specific classification requirements of this study. In addition, for BubbleNet2, the softmax layer was removed, allowing the model to perform regression tasks to predict elastic modulus values directly. The models were trained on a dataset comprising 130 scalograms and 190 B-mode as well as CEUS images for each fibrin concentration. Model performance was validated using an unseen test set, which included 25 scalograms and 30 B-mode and CEUS mode images for each fibrin concentration (Fig. 2).

Hyperparameters were fine-tuned using a Bayesian optimization technique with five-fold cross-validation to identify the optimal settings for each image type. For scalogram images, the optimized hyperparameters included a learning rate of 1.16×10^{-4} , a batch size of 40, and L2 regularization of 7.02×10^{-8} . For B-mode images, the optimal settings were a learning rate of 1.08×10^{-4} , L2 regularization of 6.67×10^{-6} , and a batch size of 52. For CEUS mode data, the optimal hyperparameters included a learning rate of 1.07×10^{-4} , L2 regularization of 2.67×10^{-5} , and a batch size of 121. Both BubbleNet1 and BubbleNet2 were trained for 20 epochs using the Adam optimizer [44], which improved convergence by incorporating first and second-order moment estimates, along with bias correction.

1) Evaluation Metrics for Classification Models: In assessing the performance of classification model, several key metrics were used to provide a comprehensive understanding of their effectiveness. We evaluated the performance of the classification model using four basic indicators of true positive, true negative, false positive, false negative, which established other quantitative metrics such as accuracy, recall, precision, specificity, and F1-score as defined in the online supplemental material.

2) Transfer Learning Implementation: A step-by-step flowchart of the proposed methodology is illustrated in Fig. 2. The transfer learning model was implemented using MATLAB, where pre-trained version of AlexNet were fine-tuned to develop BubbleNet1 and BubbleNet2. Five-fold cross-validation was employed to ensure robust training and validation of the models. The hyperparameters for training both models were fine-tuned using Bayesian validation, ensuring optimal model performance.

F. Statistical Analyses

Statistical analyses were performed in GraphPad Prism software (GraphPad Software, Inc., La Jolla, CA, USA) using one-way ANOVA followed by Tukey's multiple comparisons test. Experimental data are presented as the mean $\pm$ standard deviation. Statistical significance was determined at a confidence level of 95% ($p < 0.05$).

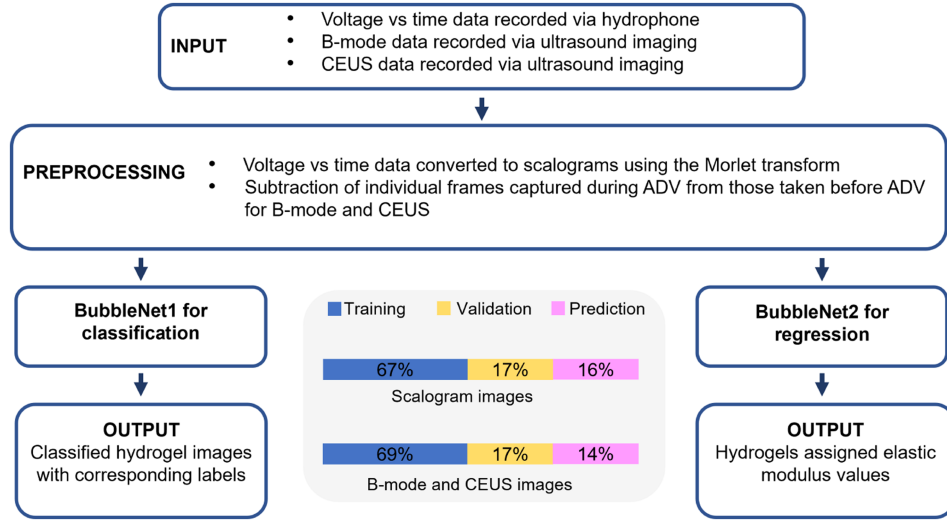


Fig. 2. The flowchart provides a step-by-step outline of the preprocessing, classification, and elastic modulus prediction for three distinct datasets obtained from hydrophone and ultrasound imaging: voltage vs. time, B-mode, and contrast-enhanced ultrasound (CEUS). The diagram also shows the allocation of acquired images for training, validation, and prediction across different imaging modalities.

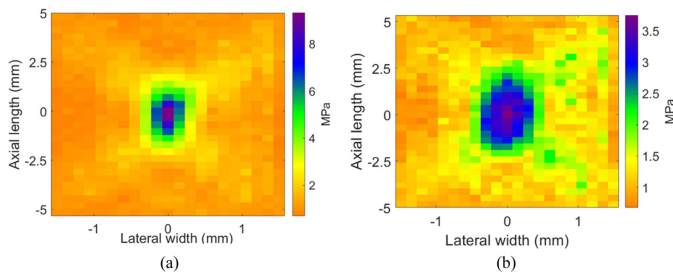


Fig. 3. The acoustic pressure distribution at the focal region of the transducer was measured, using a fiber optic hydrophone, along the axial and lateral directions to quantify the (a) peak positive and (b) peak negative pressures at 2.5 MHz.

III. RESULTS

A. Beam Characterization

The beam characterization at the transducer focus yielded an axial length of 4.7 ± 0.2 mm and a lateral width of 0.6 ± 0.2 mm at full width half maximum for 2.5 MHz and 4 MPa (Fig. 3). The axial length and lateral width suprathreshold for ADV (> 2.5 MPa) [18] were 4.2 ± 0.3 mm and 0.7 ± 0.1 mm, respectively. Due to the combined effects of nonlinear propagation and diffraction, the ratio of peak positive to peak negative was 2.6.

B. Mechanical Characterization of Fibrin-Based Hydrogels

The AFM measurements on fibrin-based tissue-mimicking hydrogels showed that the Young's modulus increased with higher fibrin concentrations, confirming the expected trend of increased stiffness with denser fibrin networks (Table I). Additionally, greater variability in Young's modulus was observed

TABLE I
MEASUREMENT OF YOUNG'S MODULUS OF FIBRIN-BASED HYDROGELS USING ATOMIC FORCE MICROSCOPY AT CONTACT MODE

Fibrin concentration (%)	Young's modulus (kPa)
0.2	1.66 ± 0.70
2	4.62 ± 1.67
4	8.79 ± 1.54

at higher fibrin concentrations, likely due to the accelerated polymerization rates, which may lead to heterogeneous crosslinking and localized stiffness variations within the hydrogels.

C. Acoustic Emissions From ADV Bubbles

Fig. 4(a)–(d) displays representative frequency-time spectrograms of the backscattered signals from ADV bubbles generated within fibrin hydrogels at varying concentrations, recorded using passive cavitation detection. The spectrograms showed that nanodroplets embedded in hydrogels with lower fibrin concentrations generated stronger backscattered signals, characterized by the presence of not only the fundamental (f_0) frequency but also subharmonic (f_0/n_i), and higher harmonic ($n_i f_0$) frequencies, a characteristic of nonlinear bubble oscillations. Time-dependent backscattered signals were recorded over 500 ms period for each experimental condition. In media with zero elasticity (i.e., DI water), significantly higher frequency components were observed compared to fibrin hydrogels. As the Young's modulus of the surrounding medium increased by ≈ 7 kPa, the subharmonic (Fig. 4(e)), fundamental (Fig. 4(f)), ultraharmonic (Fig. 4(g)), and second harmonic (Fig. 4(h)) components of the backscattered response decreased by 25 dB, 15 dB, 20 dB, and 25 dB, respectively. Acoustic emissions were sampled every 10 ms over the 500 ms window, capturing the sustained presence and oscillatory behavior of ADV bubbles

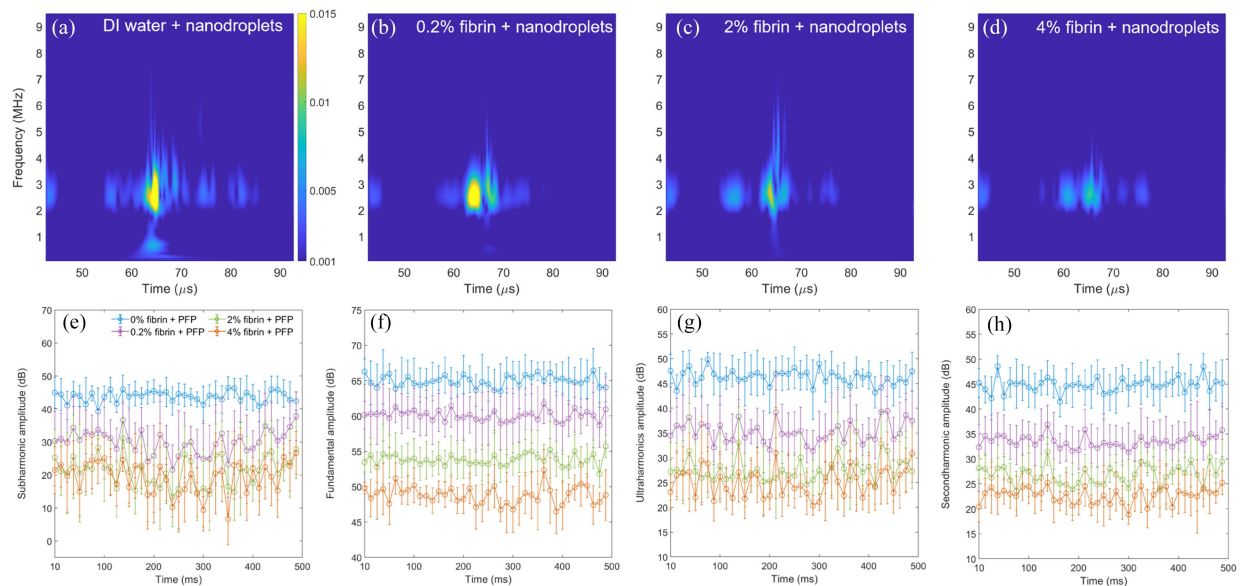


Fig. 4. Acoustic emissions from ADV-generated bubbles in fibrin hydrogels were recorded using passive cavitation detection. Frequency-time power spectra of the recorded signals are compared for (a) 0% (i.e., DI water), (b) 0.2%, (c) 2%, and (d) 4% fibrin hydrogels. The temporal evolution of acoustic emissions from ADV bubbles is presented, highlighting changes in (e) subharmonic amplitude, (f) fundamental amplitude, (g) ultraharmonic amplitude, and (h) second harmonic amplitude over 500 ms for fibrin concentrations of 0% (blue), 0.2% (purple), 2% (green), and 4% (orange).

during ultrasound exposure. Fig. 4(e)–(h) also indicate that while stable harmonic generation was observed, subsequent ultrasound pulses did not produce additional bubbles, suggesting that the majority of nanodroplets within the focal region had already vaporized. This finding underscores the efficiency of ADV in producing consistent acoustic emissions, dependent on the mechanical properties of the surrounding hydrogel matrix.

ADV bubbles were further visualized as echogenic bubble clouds within fibrin hydrogels of varying concentrations using differential B-mode and CEUS ultrasound imaging, where the reference frame was subtracted from the ADV frames (Fig. 5). B-mode images indicated that the axial length of the ADV bubble cloud decreased significantly with increasing fibrin concentration, suggesting that stiffer hydrogels restricted bubble growth. However, the lateral width of the bubbles only showed a significant reduction at the highest fibrin concentration. Comparing with suprathreshold ultrasound beam dimensions, the axial length of the ADV bubble cloud in DI water (3.7 ± 0.7 mm) was not significantly different from the suprathreshold axial beam length (i.e., 4.2 ± 0.3 mm). In contrast, the axial lengths of the ADV bubble clouds in fibrin hydrogels were significantly reduced, measuring 2.8 ± 0.3 mm in 0.2% fibrin, 1.9 ± 0.2 mm in 2% fibrin, and 1.6 ± 0.2 mm in 4% fibrin hydrogels. Interestingly, the lateral widths of the ADV bubble clouds across all conditions- 1.3 ± 0.03 mm (DI water), 1.4 ± 0.09 mm (0.2% fibrin), 1.3 ± 0.08 mm (2% fibrin), and 1.0 ± 0.07 mm (4% fibrin)- were significantly greater than the suprathreshold beam width at the focus (i.e., 0.7 ± 0.1 mm). For CEUS images, the average axial lengths of the ADV bubble clouds were 2.4 ± 0.08 mm, 2.1 ± 0.2 mm, 1.7 ± 0.2 mm, and 1.2 ± 0.2 mm in DI water, 0.2%, 2%, and 4% fibrin, respectively. The lateral widths of the ADV bubbles were 1.4 ± 0.08 mm, 1.7 ± 0.03 mm, $1.0 \pm$

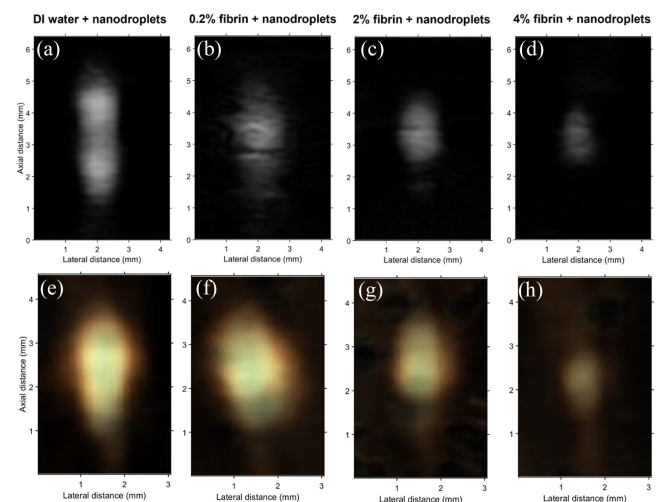


Fig. 5. Acoustic emissions from bubbles generated in fibrin hydrogels via ADV were visualized using B-mode and nonlinear CEUS ultrasound imaging. Representative B-mode images of ADV bubble clouds are shown for fibrin concentrations of (a) 0%, (b) 0.2%, (c) 2%, and (d) 4%, illustrating the impact of fibrin concentration on imaging characteristics of ADV bubble clouds. Similarly, CEUS images are given for fibrin concentrations of (e) 0%, (f) 0.2%, (g) 2%, and (h) 4%.

0.1 mm, and 0.8 ± 0.06 mm in DI water, 0.2%, 2%, and 4% fibrin, respectively. It should be noted that no ground truth data, such as optical imaging, were collected to validate the dimensions of the ADV bubble cloud.

Ultrasound images of ADV bubbles in fibrin hydrogels were further processed to extract the corresponding morphological as well as textural features. The GLCMs were calculated at four directions— 0° , 45° , 90° , 135° —with a distance parameter of

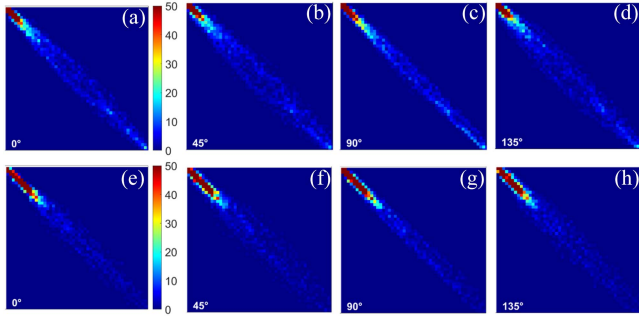


Fig. 6. The corresponding GLCMs for B-mode images of ADV bubble clouds were computed to highlight textural differences and spatial relationships. GLCMs are shown for ADV bubble clouds in DI water at (a) 0°, (b) 45°, (c) 90°, and (d) 135°, and in 4% fibrin hydrogels at (e) 0°, (f) 45°, (g) 90°, and (h) 135°.

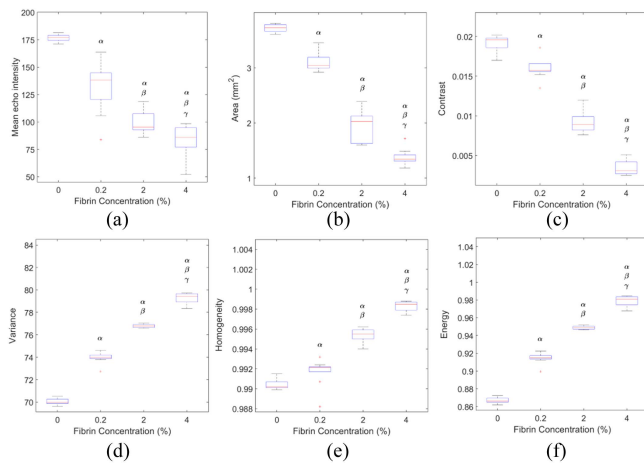


Fig. 7. B-mode images were further analyzed to compare (a) mean pixel intensity, (b) ADV bubble cloud area, (c) contrast, (d) variance, (e) homogeneity, and (f) energy. Statistically significant differences ($p < 0.05$) are denoted as follows: α : vs. 0%; β : vs. 0.2%; γ : vs. 2%.

1 to assess for any preferred directionality in texture analysis (Fig. 6). The parameters did not significantly differ across directions (Fig. S1). Therefore, we plotted the average of the GLCM parameters from all four angles to compare ADV-bubble behavior in different fibrin concentrations. For first-order texture analysis, two key features were calculated and compared among fibrin concentrations: mean echo intensity (Fig. 7(a)) and ADV bubble cloud area (Fig. 7(b)). The mean echo intensity of ADV bubble clouds in DI water (i.e., 0% fibrin) was approximately three times higher than in 4% fibrin. The obtained ADV bubble cloud areas, presented in Fig. 7(b), correlated inversely with fibrin concentration. Fig. 7(c)–(f) display the textural features and quantitative parameters derived from GLCM analysis of B-mode images for different fibrin concentrations (0%, 0.2%, 2%, and 4%), corresponding to different Young's moduli (Table I). As fibrin concentration increased, the contrast of ADV bubble clouds decreased (Fig. 7(c)), indicating less variation in texture and local pixel intensities and suggesting that bubble clouds become more homogeneous at higher fibrin densities. Other GLCM parameters, such as variance (Fig. 7(d)), homogeneity

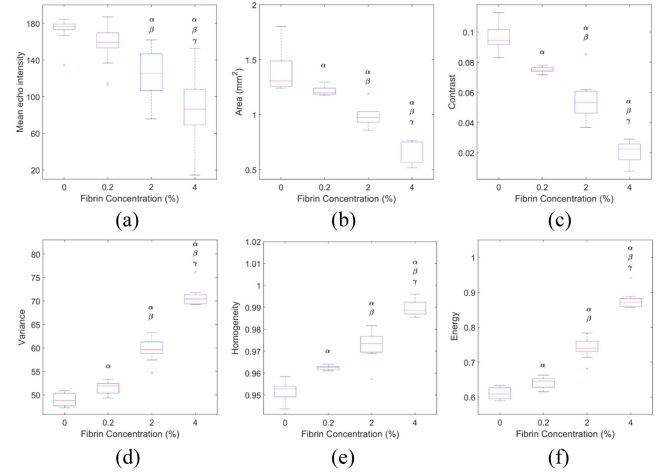


Fig. 8. Box plots showing quantitative parameters obtained from GLCM analysis and pixel intensity measurements of CEUS images for varying fibrin concentrations (0%, 0.2%, 2%, and 4%). The features include (a) mean echo intensity, (b) ADV bubble cloud area, (c) contrast, (d) variance, (e) homogeneity, and (f) energy. Statistically significant differences ($p < 0.05$) are denoted as follows: α : vs. 0%; β : vs. 0.2%; γ : vs. 2%.

(Fig. 7(e)), and energy (Fig. 7(f)), correlated directly with fibrin density. The rise in homogeneity indicates that pixel intensities within the B-mode images are more uniform and closely related, leading to smoother texture patterns. Similarly, the increase in energy, which measures the sum of squared elements in the GLCM, further supports the enhanced textural uniformity observed in stiffer hydrogels.

Fig. 8 presents the texture features and quantitative parameters derived from CEUS images across different fibrin concentrations. The trends observed in these metrics were consistent with those obtained from B-mode images. Specifically, mean pixel intensity (Fig. 8(a)) and ADV bubble cloud area (Fig. 8(b)) both decreased at higher fibrin concentrations. Additionally, contrast (Fig. 8(c)) diminished at higher fibrin densities, while variance (Fig. 8(d)), homogeneity (Fig. 8(e)), and energy (Fig. 8(f)) all increased with higher fibrin concentrations. These results emphasize the consistent behavior of ADV-generated bubbles in fibrin hydrogels with varying Young's moduli, as detected through both B-mode and CEUS imaging modalities. Among the six features analyzed here, bubble cloud area, contrast, and variance demonstrated greater differentiation among the four fibrin concentrations, with small standard deviations and significant differences observed between groups. The correlation matrices among the six analyzed features are shown in Fig. 9. Higher mean echo intensity was associated with larger ADV bubble cloud area, increased contrast (i.e., higher gray level dispersion), and lower homogeneity.

D. Transfer Learning

Fig. 10 provides a detailed evaluation of the transfer learning model's training and validation performance across three data

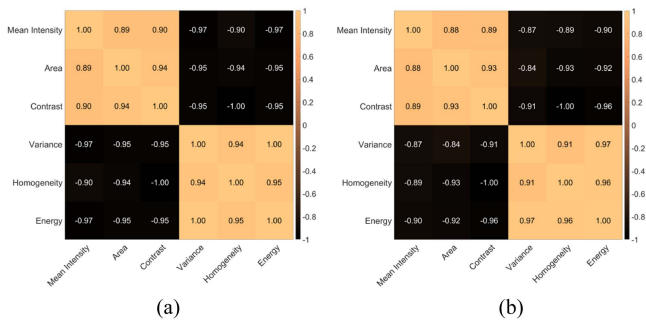


Fig. 9. Correlation matrix between GLCM parameters, mean echo intensity, and ADV bubble cloud area for (a) B-mode and (b) CEUS images.

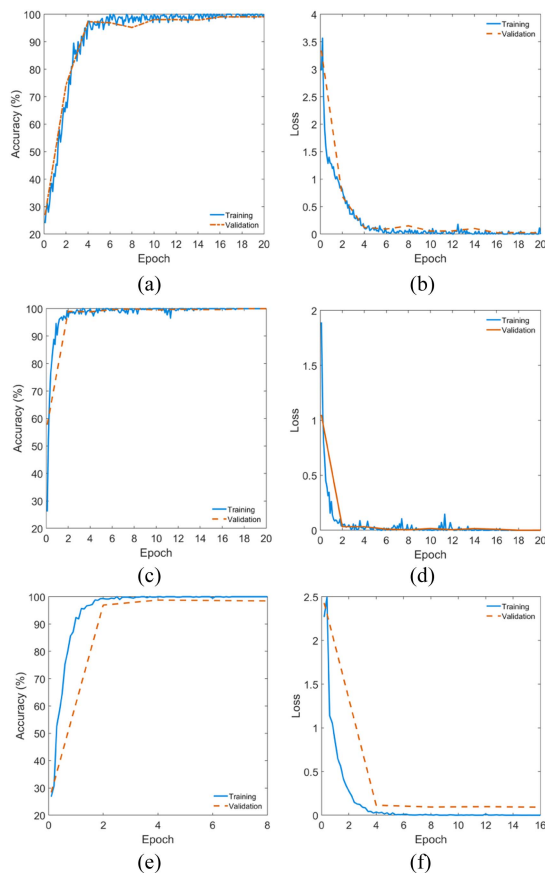


Fig. 10. Model performance evaluation using five-fold cross-validation. The average accuracy and loss are presented for each data modality: (a) accuracy and (b) loss for scalogram images, (c) accuracy and (d) loss for B-mode images, and (e) accuracy and (f) loss for CEUS images, highlighting the model's training and validation performance across different data types.

modalities: scalogram, B-mode, and CEUS images. Performance was assessed using five-fold cross-validation, ensuring a robust evaluation that minimized the influence of any single subset of data on the model's overall performance. During the initial epochs, both training and validation accuracy increased rapidly across all modalities, reflecting the model's ability to

learn effectively from the provided data. Similarly, the loss plots showed a sharp decline during early stages of training, stabilizing as training progressed, indicating successful convergence. For scalogram images, the model achieved a peak average training accuracy of 99.8% and a validation accuracy of 99.0% by the final epoch (Fig. 10(a)). Corresponding loss values were low, with 0.025 for the training set and 0.028 for the validation set (Fig. 10(b)). For B-mode images, a training accuracy of 100% and a validation accuracy of 99.7%, with a final average loss of 6×10^{-4} for the training set and 5.7×10^{-4} for the validation set (Fig. 10(c)–(d)) were obtained. In the case of CEUS data, the model achieved 100% training accuracy and 99.5% validation accuracy (Fig. 10(e)). The average loss values were 0.0007 for the training set and 0.092 for the validation set, as depicted in Fig. 10(f).

Fig. 11 presents confusion matrices for classifying fibrin concentrations using scalogram (Fig. 11(a)), B-mode (Fig. 11(b)), and CEUS images (Fig. 11(c)). The confusion matrices provide a detailed numerical comparison between the model's predictions and the actual data. For scalogram images, the model achieved perfect classification accuracy for 0% and 0.2% fibrin concentrations, while maintaining high accuracy of 92% for 2% and 4% fibrin concentrations. Similarly, for B-mode data, the model attained perfect classification accuracy for 0%, 0.2%, and 4% fibrin concentrations (100%), and a near-perfect accuracy of 96.67% for the 2% fibrin concentration. CEUS images resulted in perfect classification among all four fibrin concentrations, indicating that CEUS data might be more sensitive to variations in fibrin concentration compared to scalogram and B-mode images.

Figure S2 displays Receiver Operating Characteristic (ROC) curves for classifying fibrin concentrations using scalogram, B-mode, and CEUS mode images, all evaluated on the unseen datasets. Each ROC curve illustrates the model's classification performance by plotting the true positive rate (sensitivity) against the false positive rate (specificity). The area under the curve (AUC) scores (Fig. S2), displayed within each plot, quantified the overall performance of the models, with higher AUC values reflecting superior discrimination power between different fibrin concentrations. Tables S1-3 provide the detailed classification reports for scalogram, B-mode, and CEUS data, respectively. These tables compare key metrics defined in Section E.

Fig. 12 illustrates the regression performance of the transfer learning model in predicting the elastic modulus of fibrin hydrogels across three different data modalities. For scalogram images, the root mean square error (RMSE) (Fig. 12(a)) for both training and validation sets dropped sharply from approximately 8.1 to a stable value around 0.26, while the corresponding loss (Fig. 12(b)) decreased from 70 to less than 0.2. Similarly, for B-mode data, the RMSE (Fig. 12(c)) declined from about 3.6 to approximately 0.23, and the loss (Fig. 12(d)) reduced from 15 to around 0.17. For CEUS images, the RMSE reduced from about 5 to 0.25 (Fig. 12(e)), and the loss dropped from 35 to below 0.3 (Fig. 12(f)). Fig. 13 compares the predicted Young's modulus values obtained from the BubbleNet2 model with the

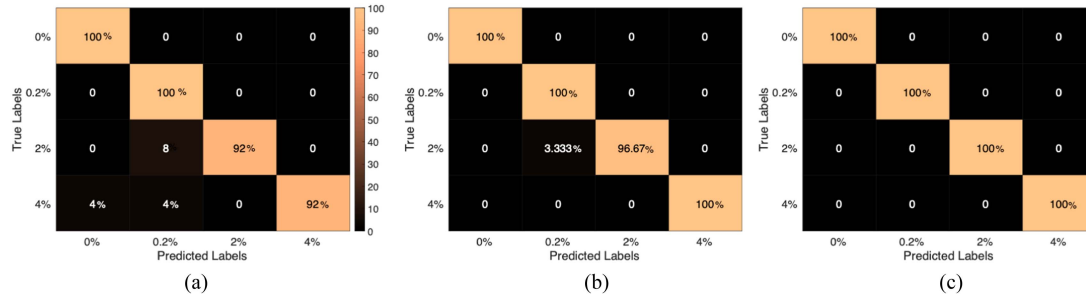


Fig. 11. Confusion matrices for classifying fibrin concentration in hydrogels using (a) Scalogram, (b) B-mode, and (c) CEUS data. Each matrix was evaluated on 25 unseen scalogram images, 30 unseen B-mode and CEUS datasets per fibrin concentration, demonstrating the model's classification performance across the various data types.

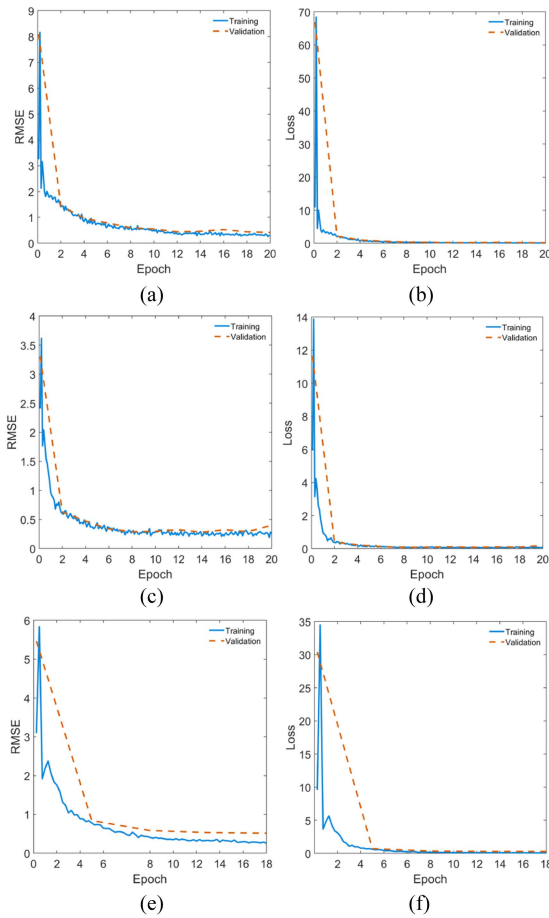


Fig. 12. Transfer learning regression results for predicting the elastic modulus values of fibrin hydrogels, evaluated using five-fold cross-validation. Root Mean Square Error (RMSE), illustrating the reduction in prediction error over training epochs, is shown for (a) scalogram, (c) B-mode, and (e) CEUS images. Corresponding loss versus epoch plots for (b) Scalogram, (d) B-mode, and (f) CEUS images demonstrate the model's effectiveness in minimizing error during training.

experimentally measured values from AFM. The predicted Young's moduli, using scalogram images, were 0.02 ± 0.1 kPa, 1.4 ± 0.03 kPa, 4.3 ± 0.8 kPa, and 8.0 ± 1.3 kPa in 0%, 0.2%, 2%, and 4% fibrin hydrogels, respectively. For B-mode image,

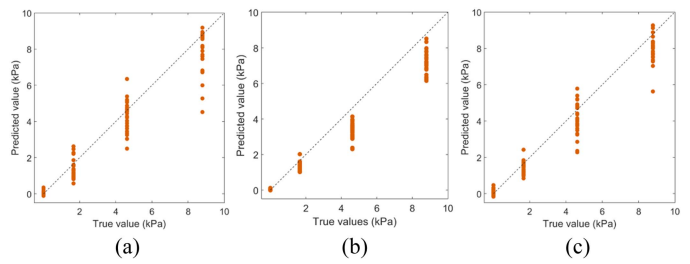


Fig. 13. Scatter plots comparing predicted elastic modulus values using BubbleNet2 versus true elastic moduli measured via AFM for (a) scalogram, (b) B-mode, and (c) CEUS images.

the predicted Young's moduli were 0.01 ± 0.03 kPa, 1.3 ± 0.2 kPa, 3.3 ± 0.4 kPa, and 7.2 ± 0.6 kPa in 0%, 0.2%, 2%, and 4% fibrin hydrogels, respectively. For CEUS image, the predicted Young's moduli were 0.08 ± 0.1 kPa, 1.4 ± 0.3 kPa, 4.1 ± 0.8 kPa, and 8.2 ± 0.7 kPa in 0%, 0.2%, 2%, and 4% fibrin hydrogels, respectively

IV. DISCUSSION

This work builds on our previous studies, which demonstrated that ADV bubbles are highly sensitive to the elastic modulus of their surrounding medium [18], [22]. Specifically, we showed that ADV bubbles respond dynamically to the mechanical properties of their environment through a combination of numerical modeling, ultra-high-speed imaging, and acoustic measurements. Numerical simulations integrating the Keller-Miksis equation with a Kelvin-Voigt model predicted reduced oscillation amplitudes and decreased nonlinearity in stiffer media. Experimental high-speed microscopy showed that increasing the fibrin concentration from 0.2% to 4%, corresponding to ≈ 5.8 kPa increase in the elastic modulus of fibrin hydrogels, resulted in a 40% reduction in maximum expansion ratio and a 53% decrease in bubble collapse time [18]. Passive cavitation detection further confirmed that both temporal and spectral responses of ADV bubbles were significantly reduced in stiffer fibrin hydrogels [22], underscoring the influence of mechanical properties on bubble dynamics and acoustic emissions. In this study, we extended our acoustic interrogations to include B-mode and nonlinear CEUS imaging characteristics

of ADV-generated bubbles in fibrin-based tissue mimicking hydrogels. To quantitatively characterize these images across various fibrin concentrations, we employed both morphological and textural analysis techniques. The application of GLCM metrics for tissue characterization through B-mode images is well established [38], [45], but their use in analyzing ADV bubble clouds in this context is new, allowing us to explore how fibrin concentration affects image texture due to ADV-bubble generation.

Our AFM measurements aligned with previous bulk rheology data [18], [22], [46], demonstrating a direct correlation between fibrin concentration and the elastic properties of fibrin hydrogels. The shear elastic modulus (G) and Young's modulus (E) are expected to follow the Trouton ratio ($E = 3G$), assuming a Poisson's ratio of 0.5 for incompressible materials. Interestingly, we observed that the Trouton ratio approached 3 only at higher fibrin concentrations. This deviation from this ratio in softer hydrogels has been reported in other studies and suggests that as fibrinogen concentration increases, the hydrogels become more incompressible [47].

In both B-mode and CEUS images, ADV bubbles in lower fibrin concentrations produced higher mean echo intensities and larger bubble cloud areas. This behavior is likely due to the larger volumetric expansion of phase-shift droplets in softer hydrogels with lower Young's moduli. Theoretical predictions support this observation; smaller microbubbles absorb more and scatter less at the fundamental frequency, while larger bubbles scatter more efficiently [48]. The scattering cross-section of a gas bubble, defined as the ratio of power scattered by the bubble to incident acoustic intensity [49], directly reflects the scattering efficiency. In ultrasound imaging, the scattering efficiency is directly related to the echogenicity of the bubbles. Since the scattering cross-section depends on bubble size, the echogenicity is similarly influenced by the size of the bubble, resulting in brighter and more heterogeneous textures in softer hydrogels. These findings align with studies in other biological contexts, such as chronic kidney disease, where increased homogeneity and reduced contrast have been associated with a decrease in fibrous and adipose tissues within muscles [50]. In our study, the GLCM parameters did not differ significantly across different directions. These findings are consistent with previous studies demonstrating that in homogeneous tissue, GLCM features are similar across different directions. In contrast, in anisotropic tissues like tendons, GLCM features vary with direction, highlighting the importance of considering directionality in such cases [37], [51], [52].

The incorporation of deep learning techniques in clinical imaging has transformed the field by enabling the extraction of complex features and the recognition of intricate patterns within medical images. Convolutional neural networks, a cornerstone of deep learning, are particularly effective in these tasks; however, their performance heavily relies on access to large, well-annotated datasets, which are often limited in medical imaging. To address the limitations posed by smaller datasets, transfer learning has emerged as a robust strategy [41], [53]. This method involves pretraining on extensive datasets such as ImageNet and

fine-tuning the network on more specialized, smaller medical datasets. By leveraging generalized low-level features in the initial layers of the network and adapting the deeper layers to focus on application-specific patterns, transfer learning can significantly improve performance in medical imaging tasks. Previous studies have successfully applied transfer learning in various medical imaging tasks. For instance, Byra et al. [54] utilized a pretrained Visual Geometry Group (VGG)-19 model to classify breast masses by converting gray scale B-mode images to RGB images. Tanaka et al. [55] developed an ensemble transfer learning model combining VGG-19 and ResNet-152 to classify breast ultrasound images as benign or malignant. Additionally, deep learning models have also been used to distinguish benign from malignant breast masses using shear wave elastography images [42].

We demonstrated the feasibility of integrating deep learning models with ADV for tissue characterization. By leveraging pretrained networks and fine-tuning them on specialized, smaller datasets, our models achieved high classification and regression performance. However, larger studies are required to validate these findings across a broader spectrum of hydrogel types and conditions. This study investigated three different fibrin concentrations to establish feasibility. We only used three fibrin concentrations due to challenges with heterogeneous polymerization at concentrations above 4%. Future work will explore alternative biomaterials with a broader range of elastic moduli, such as gelatin, polyacrylamide, and polyethylene glycol. Furthermore, all experiments in this study were conducted using tissue-mimicking hydrogels. While these provide controlled environments for testing, they do not fully capture the complexities and heterogeneity of biological tissues and may not be representative of ADV bubble cloud formation in vivo. Ultrasound images were also only acquired in a fixed 2D plane, whereas a volume of bubble cloud was generated. Additionally, the sensitivity of ADV-bubbles to the mechanical properties of surrounding media may be further enhanced by utilizing higher frequencies, a direction currently under investigation.

V. CONCLUSION

ADV is a promising platform for imaging and therapeutic applications of ultrasound. By integrating both passive and active acoustic detection methods, we characterized the acoustic emissions of phase-shift nanodroplets embedded in fibrin-based hydrogels of varying elastic moduli. The results revealed that ADV bubbles produced significantly stronger backscattered signals in softer hydrogels, highlighting their sensitivity to changes in the surrounding medium. We further quantified the morphological and textural characteristics, using GLCM analysis, providing valuable insights into the relationship between fibrin concentration and acoustic and imaging characteristics of ADV bubble clouds. The use of transfer learning with convolutional neural networks further enhanced the diagnostic potential of ADV by allowing for accurate classification and regression tasks, underscoring the feasibility of integrating deep learning

frameworks and ADV. Overall, this study presents a novel approach that combines ADV, quantitative ultrasound, and deep learning to expand the applications of ADV.

DISCLOSURES

The authors declare no conflicts of interest.

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